

EX. NO. 9**MONITORING AND LOGGING FABRIC NETWORKS****AIM**

To monitor the Hyperledger Fabric network by inspecting Docker containers, viewing peer and orderer logs, checking network status, monitoring resource utilization, and analyzing blockchain transaction logs for troubleshooting and performance analysis.

SOFTWARE REQUIREMENTS

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

Stop any existing Fabric network and start a fresh network with the application channel and Certificate Authorities.

```
./network.sh down
```

./network.sh up createChannel -ca

```
vboxuser@Ubuntu: ~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network
vboxuser@Ubuntu: ~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Using docker and docker compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'Certificate Authorities'
Bringing up network
```

Step 3: Verify Running Containers

Display all active Docker containers. docker

ps

```
vboxuser@Ubuntu: ~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                            CREATED        STATUS
PORTS
2c5baef255f3   hyperledger/fabric-peer:latest      "peer node start"                               31 seconds ago Up
0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
a44b18b6714e   hyperledger/fabric-orderer:latest   "orderer"                                         31 seconds ago Up
0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
ab1b97191d8a   hyperledger/fabric-peer:latest      "peer node start"                               31 seconds ago Up
0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
14b030542e7e   hyperledger/fabric-ca:latest         "sh -c 'fabric-ca-se..."                       45 seconds ago Up
0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp
818d5fea0349   hyperledger/fabric-ca:latest         "sh -c 'fabric-ca-se..."                       45 seconds ago Up
0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp, 7054/tcp, 0.0.0.0:19054->19054/tcp, [::]:19054->19054/tcp
9d18e23bd759   hyperledger/fabric-ca:latest         "sh -c 'fabric-ca-se..."                       45 seconds ago Up
0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp, 7054/tcp, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp
```

Step 4: Monitor CPU and Memory Usage

Display live CPU and memory utilization of the Fabric containers.

docker stats

Allow the statistics to run for a few seconds and press Ctrl+C to stop.

For a single snapshot: docker

stats --no-stream

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O
2c5baef255f3	peer0.org2.example.com	7.26%	26.55MiB / 5.625GiB	0.46%	225kB / 154k
B	45.1kB / 393kB	9			
a44b18b6714e	orderer.example.com	0.39%	16.38MiB / 5.625GiB	0.28%	60.6kB / 207
kB	5.1MB / 193kB	8			
ab1b97191d8a	peer0.org1.example.com	8.96%	30.62MiB / 5.625GiB	0.53%	225kB / 154k
B	0B / 393kB	8			
14b030542e7e	ca_org1	0.00%	9.527MiB / 5.625GiB	0.17%	40.1kB / 50k
B	0B / 737kB	7			
818d5fea0349	ca_orderer	0.00%	11.05MiB / 5.625GiB	0.19%	60.5kB / 84.
6kB	999kB / 1.05MB	7			
9d18e23bd759	ca_org2	0.00%	9.633MiB / 5.625GiB	0.17%	37.9kB / 44k
B	1.16MB / 737kB	7			

Step 5: View Peer Logs

Display the recent logs of the Org1 peer.

```
docker logs --tail 50 peer0.org1.example.com
```

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs --tail 50 peer0.org1.example.com
2026-08-31 16:48:12.384 UTC 001c INFO [discovery] NewService -> Created with config TLS: true,
authCacheMaxSize: 1000, authCachePurgeRatio: 0.750000
2026-08-31 16:48:12.384 UTC 001d INFO [nodeCmd] serve -> Discovery service activated
2026-08-31 16:48:12.384 UTC 001e INFO [nodeCmd] serve -> Starting peer with Gateway enabled
2026-08-31 16:48:12.384 UTC 001f INFO [nodeCmd] serve -> Starting peer with ID=[peer0.org1.example.com], network ID=[dev], address=[peer0.org1.example.com:7051]
2026-08-31 16:48:12.386 UTC 0020 INFO [nodeCmd] serve -> Started peer with ID=[peer0.org1.example.com], network ID=[dev], address=[peer0.org1.example.com:7051]
2026-08-31 16:48:12.387 UTC 0021 INFO [kvledger] LoadPreResetHeight -> Loading prereset height from path [/var/hyperledger/production/ledgersData/chains]
2026-08-31 16:48:12.387 UTC 0022 INFO [blkstorage] preResetHtFiles -> No active channels passed
2026-08-31 16:48:18.742 UTC 0023 INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel] with genesis block
2026-08-31 16:48:18.783 UTC 0024 INFO [blkstorage] newBlockfileMgr -> Getting block information
```

Step 6: View Orderer Logs

Display the recent logs of the ordering service.

```
docker logs --tail 50 orderer.example.com
```

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs --tail 50 orderer.example.com
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
2026-08-31 16:48:11.755 UTC 000f INFO [orderer.common.server] Main -> Beginning to serve requests
2026-08-31 16:48:15.437 UTC 0010 INFO [blkstorage] newBlockfileMgr -> Getting block information from block storage
2026-08-31 16:48:15.466 UTC 0011 INFO [orderer.consensus.etcdraft] HandleChain -> EvictionSuspicion not set, defaulting to 10m0s
2026-08-31 16:48:15.466 UTC 0012 INFO [orderer.consensus.etcdraft] HandleChain -> Without system channel: after eviction Registrar.SwitchToFollower will be called
2026-08-31 16:48:15.467 UTC 0013 INFO [orderer.consensus.etcdraft] createOrReadWAL -> No WAL data found, creating new WAL at path '/var/hyperledger/production/orderer/etcdraft/wal/mychannel' channel=mychannel node=1
```

Step 7: View Certificate Authority Logs

Display the logs of the Org1 Certificate Authority.

```
docker logs ca_org1
```

Step 8: Verify the Network Channel

Load the Org1 peer environment and list the channels joined by the peer.

```
source setOrg1.sh peer
```

```
channel list
```

Step 9: Deploy the Asset Transfer Chaincode

Deploy the JavaScript Asset Transfer chaincode to mychannel.

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
```

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-08-31 16:51:48.475 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ source setOrg1.sh
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Using docker and docker compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
```


Step 10: Query Installed Chaincode

Verify that the chaincode package has been installed on the peer.

```
peer lifecycle chaincode queryinstalled
```

Step 11: Set TLS Environment Variables

Set the TLS certificates for Org1 and Org2.

```
export
```

```
ORG1_TLS=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
```

```
export
```

```
ORG2_TLS=${PWD}/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt
```

A terminal window with a dark background and light-colored text. The first line shows a message: 'Chaincode initialization is not required'. The second line shows a command: 'vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network\$ peer lifecycle chaincode queryinstalled'. The third line shows the output: 'Installed chaincodes on peer:'. The fourth line shows the package ID and label: 'Package ID: basic_1.0:1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629, Label: basic_1.0'. The fifth line shows the export command for ORG1_TLS: 'vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network\$ export ORG1_TLS=\${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt'. The sixth line shows the export command for ORG2_TLS: 'vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network\$ export ORG2_TLS=\${PWD}/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt'.**Step 12: Initialize the Ledger**

Initialize the ledger with the sample assets. Both Org1 and Org2 peers are included for endorsement.

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride  
orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses  
localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 -  
tlsRootCertFiles $ORG2_TLS -c '{"function": "InitLedger", "Args": []}'
```

Step 13: Monitor Blockchain Transactions

Query all assets stored in the ledger. peer chaincode query -C mychannel -n

```
basic -c '{"Args":["GetAllAssets"]}'
```

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:7051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"InitLedger","Args":[]}'
2026-08-31 16:53:30.660 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
```

Step 14: View Recent Docker Events

Monitor Docker events generated by the Fabric network during the previous ten minutes.

docker events --since 10m

Press Ctrl+C after capturing the required events.

For a fixed monitoring period: timeout

15 docker events --since 10m

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ timeout 15 docker events --since 10m
2026-08-31T16:47:52.574800210Z volume create d07763cff4f20fcee772a4c97f51188990dfa41589560306b47535e5abecf1bc (driver=local)
2026-08-31T16:47:52.578867986Z volume create aef332f7adc6ef0b0db7cf7d7899ea9a94a7403a9de15b011e28935df1a590f9 (driver=local)
2026-08-31T16:47:52.619590321Z container create 5351fdef77e8a42e226d7c9e7ccb30d45c1db165e69f6b423c02cc9002557038 (image=hyperledger/fabric-peer:latest, name=hardcore_torvalds, org.opencontainers.image.created=2026-06-17T22:30:08.458Z, org.opencontainers.image.description=Hyperledger Fabric is an enterprise-grade permissioned distributed ledger framework for developing solutions and applications. Its modular and versatile design satisfies a broad range of industry use cases. It offers a unique approach to consensus that enables performance at scale while preserving privacy., org.opencontainers.image.licenses=Apache-2.0, org.opencontainers.image.revision=f871cf92a026aba7b12e6f06d71ded3e6e659d71, org.opencontainers.image.source=https://github.com/hyperledger/fabric, org.opencontainers.image.title=fabric, org.opencontainers.image.url=https://github.com/hyperledger/fabric, org.opencontainers.image.version=v2.5.16)
```

Step 15: Stop the Fabric Network

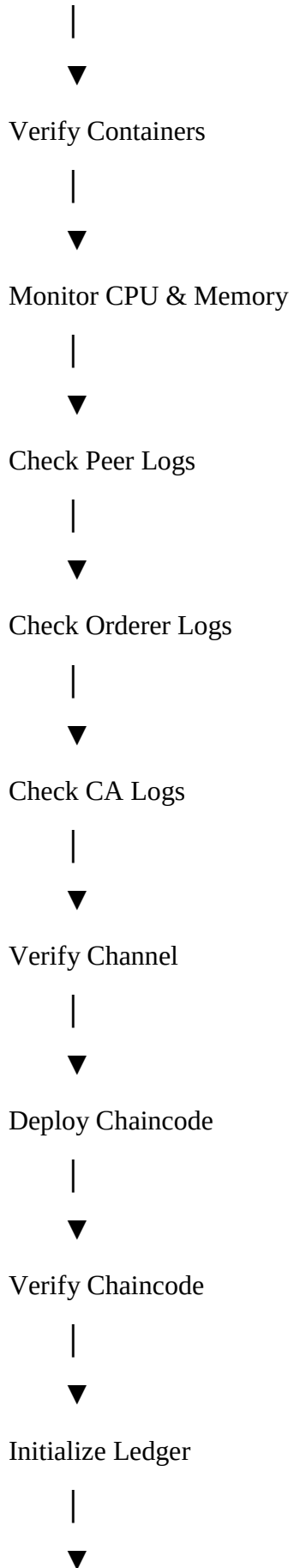
Stop and remove the Fabric network.

./network.sh down

```
~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker compose
Stopping network
[+] down 1/6
 : Container ca_org1          Stopping          1.3s
 : Container ca_org2          Stopping          1.3s
 : Container peer0.org1.example.com Stopping          1.3s
 ✓ Container orderer.example.com Removed          1.1s
 : Container peer0.org2.example.com Stopping          1.3s
 : Container ca_orderer       Stopping          1.3s
```

MONITORING WORKFLOW

Start Fabric Network



Query Ledger

|



Monitor Docker Events

|



Stop Fabric Network

RESULT

The Hyperledger Fabric network was successfully monitored using Docker utilities and Fabric CLI commands. Peer nodes, the ordering service, Certificate Authorities, channels, chaincode installation, and blockchain ledger data were verified through monitoring commands and log analysis. The experiment demonstrated effective monitoring and troubleshooting techniques for maintaining a reliable Hyperledger Fabric network.